BOARD EXAM PAPERS AND KEY

Class 10 SSLC Physics - Previous 5 Years Board Question Papers with Answer Key

Current Electricity, Electromagnetic Induction, Light Reflection & Dispersion of Light

Prescribed Institution / Board:	General Education Department & Central Board (SSLC / CBSE)
Degree / Academic Level:	Class 10 (Secondary School Level)
Subject / Module Code:	Physics (Science Paper 1)
Curriculum Academic Year:	2026-2027
Author / Academic Department:	SCERT Kerala & State Examination Directorate
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Part 1: Current Electricity & Heating Effects (Solved Board Questions)

Question 1: Joules Law of Heating (4 Marks)

State Joules law of heating and derive the relationship between heat produced (H), resistance (R), current (I), and time (t).

Solution: Joules Law states that the heat produced in a resistor is directly proportional to: (1) square of current (I^2), (2) resistance (R), (3) time of flow (t). Formula: H = I^2 * R * t = V * I * t = (V^2 / R) * t.

Question 2: Resistors in Series and Parallel Comparison

Series Combination: R_total = R1 + R2 + R3. Current remains identical across all resistors, while voltage divides.

Parallel Combination: 1 / R_total = (1/R1) + (1/R2) + (1/R3). Voltage across each branch remains constant; domestic home circuits are strictly wired in parallel.

Part 2: Electromagnetic Induction & Light

Question 3: Flemings Left & Right Hand Rules

Flemings Left Hand Rule (Motor Rule): Thumb represents Force/Motion, Forefinger represents Magnetic Field, Middle finger represents Current.

Flemings Right Hand Rule (Generator / Induced Current): Thumb represents Motion of conductor, Forefinger represents Field, Middle finger represents Direction of Induced Current.